Ai Agent Failure Modes

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Ai is a statistical generator; thus, an Ai fails where statistical models fail.

Therefore as large pre-training happens, the ai gets biased towards whichever way the conversation is most likely to continue.
Hence, any follow-up prompt depends on the previous prompt/context.

In the current scenario of Deep Cognitive Task Framework to create sophisticated, autonomous Ai agents capable of executing complex, long-horizon tasks.

My guess complex and long-horizon are the main failure properties. As most if not all of current Ai are made for short term, simple tasks such as conversation (1 time recall of all of previous conversation + simple text generation).

Failure modes are:
1. Task has enough complexity to not be broken into sub-tasks.
2. Task has an intermediate sub-task that doesn't have a specific tool.
3. Task has contradictions in its construction, so two sub-tasks are opposite to each other.
4. Task has enough sub-tasks to overflow the context management system.
5. Prompt is unable to interpreted into tasks.
6. Ai agent is unable supervise in a deterministic way because it is a statistical generator.
7. Ai agent is biased in interpretation due to not being specifically trained on possible tasks or sub-tasks.